Worksheet 44 · Similar Shapes 2 IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Use area scale factors for similar shapes • Use volume scale factors for similar solids • Convert between length, area and volume scale factors		
Grade: / 42	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Linear Scale Factor → k

Area Scale Factor → k^2

Volume Scale Factor → k^3

Questions *Similar Shapes 2*

Question 1 (3 marks)

The diagram shows two similar bottles, **A** and **B**.



Diagram **NOT** accurately drawn

Bottle **A** has surface area 240 cm²
Bottle **B** has surface area 540 cm² and volume 2025 cm³

Work out the volume of bottle **A**.

..... cm³

Question 2 (3 marks)

Solid **A** and solid **B** are mathematically similar.

Solid **A** has surface area 384 cm²
Solid **B** has surface area 864 cm²

Solid **B** has a volume of 2457 cm³

Calculate the volume of solid **A**.

..... cm³

Question 3 (3 marks)

The diagram shows two mathematically similar vases, **A** and **B**.

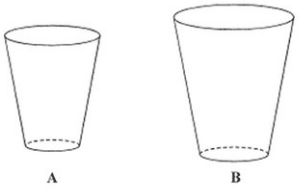


Diagram **NOT** accurately drawn

A has a volume of 405 cm³
B has a volume of 960 cm³

B has a surface area of 928 cm²

Work out the surface area of **A**.

..... cm²

Question 4 (4 marks)

The circumference of circle **B** is 90% of the circumference of circle **A**.

(a) Find the ratio of the area of circle **A** to the area of circle **B**.

..... (2)

Square **E** has sides of length e cm.
Square **F** has sides of length f cm.

The area of square **E** is 44% greater than the area of square **F**.

(b) Work out the ratio $e : f$

..... (2)

Question 5 (4 marks)

A and **B** are two similar vases.

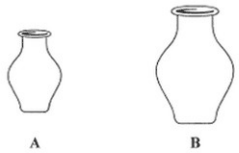


Diagram **NOT** accurately drawn

Vase **A** has height 24 cm.
Vase **B** has height 36 cm.

Vase **A** has a surface area of 960 cm^2

(a) Work out the surface area of vase **B**.

..... cm^2
(2)

Vase **B** has a volume of V cm^3

(b) Find in terms of V, an expression for the volume, in cm^3, of vase **A**.

..... cm^3
(2)

Question 6 (4 marks)

The diagram shows two similar vases, **A** and **B**.

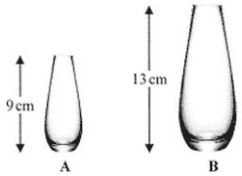


Diagram **NOT** accurately drawn

The height of vase **A** is 9 cm and the height of vase **B** is 13 cm.

Given that

surface area of vase **A** + surface area of vase **B** = 1800 cm^2

calculate the surface area of vase **A**.

..... cm^2

Question 7 (4 marks)

A frustum is made by removing a small cone from a large cone.
The cones are mathematically similar.

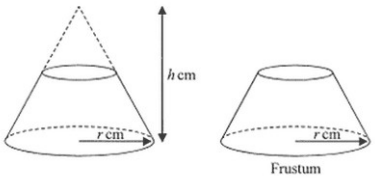


Diagram **NOT** accurately drawn

The large cone has base radius r cm and height h cm.

Given that

$$\frac{\text{volume of frustum}}{\text{volume of large cone}} = \frac{98}{125}$$

find an expression, in terms of h, for the height of the frustum.

..... cm

Question 8 (3 marks)

$ABCDE$ and $AFGHJ$ are regular pentagons.

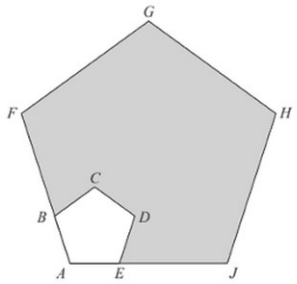


Diagram **NOT** accurately drawn

AEJ and ABF are straight lines.

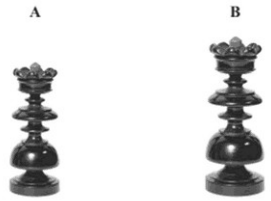
$EJ = 4AE$
The area of $ABCDE$ is 8 cm^2

Calculate the area of the shaded region.

..... cm^2

Question 9 (3 marks)

Here are two similar solid shapes.



surface area of shape **A** : surface area of shape **B** = 3 : 4

The volume of shape **B** is 10 cm^3

Work out the volume of shape **A**.
Give your answer correct to 3 significant figures.

..... cm^3

Question 10 (4 marks)

A and **B** are two similar vases.

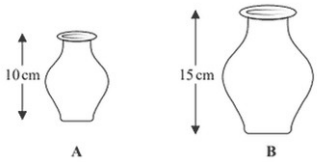


Diagram **NOT** accurately drawn

Vase **A** has height 10 cm.
Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

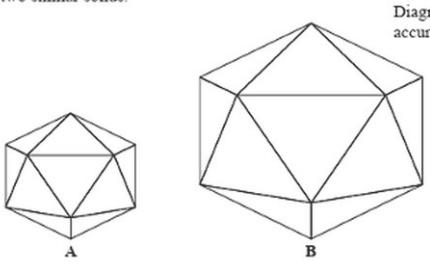
Calculate the volume of vase **A**

..... cm^3

Question 11 (3 marks)

A and **B** are two similar solids.

Diagram **NOT** accurately drawn



A has a volume of $1836\,cm^3$
B has a volume of $4352\,cm^3$

B has a total surface area of $1120\,cm^2$

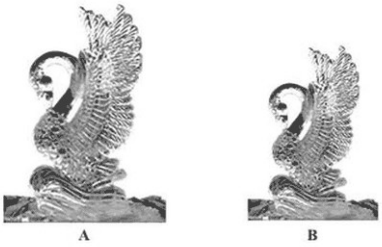
Work out the total surface area of **A**.

..... cm^2

Question 12 (4 marks)

The diagram shows two similar metal statues.

Diagram **NOT** accurately drawn



The volume of statue **B** is 20% less than the volume of statue **A**

The surface area of statue **B** is k% less than the surface area of statue **A**

Work out the value of k
Give your answer correct to 3 significant figures.

k = ...